

ULTRABOND ECO P909 2K

Two-component, solvent-free, polyurethane quick adhesive with very low emission level of volatile organic compounds, class EC1 R Plus, for all types of parquet



WHERE TO USE

Bonding of all types and formats of wooden flooring. Bonding of acoustic mats made of rubber and/or cork.

SOME APPLICATION EXAMPLES

Ultrabond Eco P909 2K is primarily used for bonding of mosaic parquet elements (EN 13488), solid lamparquet products (EN 13227), solid parquet elements with grooves and/or tongues (EN 13226), pre-assembled hardwood boards (EN 13629), multi-layer parquet elements (EN 13489) and all types of wood and pre-finished wood on cementitious screeds; **Mapecem**, **Mapecem Pronto**, **Topcem**, **Topcem Pronto** screeds, and similar products; existing flooring in ceramic tiles, marble, terrazzo tiles. Existing bonded wooden flooring, and anhydrite screeds it is also suitable for heating systems.

It may also be used to bond acoustic mats (usually composed of rubber and/or cork) to cementitious substrates or waterproofing membranes. **Ultrabond Eco P909 2K** may also be used on top of acoustic mats to bond wood and other flooring materials.

TECHNICAL CHARACTERISTICS

Ultrabond Eco P909 2K is a two-component, water- and solvent-free adhesive, with very low emission of volatile organic compounds (EMICODE EC1 R Plus). Component A is a polyurethane polymer and component B is a hardener in liquid form. By mixing the two components together, a uniform coloured product is obtained that can be easily applied with a notched trowel and has an excellent rib stability. After hardening (approximately 24 hours at room temperature) by chemical reaction, **Ultrabond Eco P909 2K** becomes a strong film with high bonding strength to all types of substrates, including those that are non absorbent.

RECOMMENDATIONS

- Before the installation of wooden flooring, always ensure that the rooms are provided with window and door frames and other suitable closures.
- The moisture of the screed must be suitable for the laying of wooden flooring in accordance with current standards.
- If the substrate is not dry or the residual moisture is above that level prescribed, it is recommended to use a suitable waterproofing primer such as **Primer MF**, **Primer MF EC Plus**, **Eco Prim PU 1K**, **Triblock P**, etc.
- If there is a danger of rising damp, always insert a vapour barrier before preparing the screed.
- Make sure that the moisture content of wood conforms to the level prescribed.
- Install at temperatures between +10°C and +30°C.
- Mix only whole packs; do not carry out partial mixtures of the adhesive.

APPLICATION PROCEDURE

Preparing the substrate

It is necessary to carefully check the substrate to verify its condition and make it suitable for installing wood.

• **Cementitious screeds:** must be cured, dry, flat and mechanically strong. The surface must be free from dust, loose parts, oil residues, paints, etc. The moisture of the screed must be measured with a carbide hygrometer. If the residual moisture is higher than required, wait until the screed is dry or apply a suitable waterproofing primer such as **Eco Prim PU 1K, Eco Prim PU 1K Turbo, Primer MF, Primer MF EC Plus, Triblock P**, etc. in order to make a vapour barrier. Substrates that are not sound enough must be removed or where possible, consolidated with a primer such as **Eco Prim PU 1K, Eco Prim PU 1K Turbo, Primer MF, Primer MF EC Plus** etc. Cracks can be repaired with products such as **Eporip, Eporip Turbo, Epojet**, etc. Particularly rough or uneven surfaces can be smoothed with levelling compounds with very high mechanical strength, such as **Ultraplan, Ultraplan Eco, Ultraplan Maxi, Nivorapid, Fiberplan**, etc. that should be chosen in relation to the thickness needed. Special MAPEI hydraulic binders can be used for the preparation of fast drying and shrinkage compensated screeds: for example, **Mapecem** or **Mapecem Pronto**, which allow installation after 1 day, **Topcem** or **Topcem Pronto** after 4 days. Screed laid onto light weight concrete or onto a grade concrete slab must be provided with a suitable vapour barrier.

• **Existing ceramic tile, marble, etc. flooring:** flooring must be clean and free from all grease, polish/paint before proceeding to bond.

• **Bonded wooden floorings:** check that the strips of existing flooring are well bonded to the substrate. Remove any paint or wax by abrasion until perfectly clean underneath wood. Vacuum clean the dust. Proceed with the next bonding.

• **Anhydrite substrates:** check and follow the instructions of the manufacturer of the screed. Always check if the screed should be sanded and primed. MAPEI recommends sanding and priming with **Eco Prim PU 1K, Eco Prim PU 1K Turbo, Primer MF and Primer MF EC Plus**. MAPEI declines all responsibilities if all the screed manufacturer's recommendations have not been followed. In case of doubt do not hesitate to contact your local MAPEI's Technical Services Department.

Mixing the adhesive

The two components of **Ultrabond Eco P909 2K** are packed in plastic containers and are already pre-measured:

- component A: 9 parts by weight, colour ochre;
- component B: 1 part by weight, colour dark brown.

Any type of modification of the ratio between component A and component B compromises the correct cross-linking of the product. Mix with a low speed stirrer until a uniform colour is obtained. Setting time and pot life strictly depends on the ambient temperature. During the summer the product hardens quickly while in winter it takes longer.

At temperatures below +10°C the reaction is too slow, therefore it is not recommended to lay wooden flooring.

Applying the product

For wood apply **Ultrabond Eco P909 2K** on the substrate using MAPEI parquet trowel for wood. The open time is approximately 1 hour under normal temperature conditions (+23°C).

To bond acoustic mats to a screed or self-levelling apply **Ultrabond Eco P909 2K** on the substrate using MAPEI no. 1 or no. 2 trowel. To bond flooring to the acoustic mat use a MAPEI trowel suitable for the flooring. The open time is approximately 1 hour under normal temperature conditions (+23°C).

Installing wood flooring

The wood that will be installed must be compliant with that prescribed by the local standards.

SET TO LIGHT FOOT TRAFFIC

The floors are set to light foot traffic after approximately 24 hours.

POLISHING

Polishing can be carried out after a minimum of 3 days, depending on the season. However, it is recommended to wait 7-10 days so the parquet can settle.

CLEANING

Whilst fresh **Ultrabond Eco P909 2K** can be removed with alcohol or specific products (**Cleaner H/Cleaner L**). Once the product has hardened, it can only be removed mechanically.

CONSUMPTION

Depends on the type and uniformity of substrate and on the flooring product. If used for wood flooring then 800-1,000 g/m² with a MAPEI parquet trowel.

If used above or below an acoustic mat then typically 350-450 g/m² with a MAPEI no.1 or no.2. or suitably-notched TKB trowel (B1 or B2).

SAFETY INSTRUCTIONS FOR PREPARATION AND APPLICATION

Ultrabond Eco P909 2K Component A is not considered as dangerous according to the current regulation regarding the classification of mixtures.

Ultrabond Eco P909 2K Component B is an irritant for the skin, eyes, and respiratory tracks. It May cause irreversible damage if used for lengthy periods and sensitisation if it comes into contact with the skin of those sensitive to isocyanates. At temperatures above +60°C the product can give off vapors that may be harmful and cause sensitization if inhaled. In case of sickness, seek medical attention. During use wear, protective clothes, gloves, and goggles protect the respiratory tracks by wearing a mask, and work in well-ventilated areas. If the product comes into contact with the eyes or skin, wash immediately with plenty of clean water and seek medical attention.

For further and complete information about the safe use of our product please refer to our latest version of the Safety Datasheet.

TECHNICAL DATA (typical values)		
PRODUCT IDENTIFICATION		
	Component A	Component B
Consistency:	paste	liquid
Colour:	ochre	dark brown
Density (g/cm³):	1.65	1.20
Dry solids content (%):	100	100
Brookfield Viscosity (mPa·s):	60,000 7 shaft - rpm 50)	20 (1 shaft - rpm 50)
EMICODE:	EC1 R Plus - very low emission	
APPLICATION DATA (at +23°C and 50% R.H.)		
Mix ratio:	A : B = 9 : 1	
Viscosity of the mix (mPa·s):	32,000 ± 4,000 (7 shaft - rpm 50)	
Density of mix (g/cm³):	1.55 ± 0.50	
Pot life of mix:	40-50 minutes	
Application temperature range:	from +10°C to +30°C	
Open time:	60 minutes	
Adjustment time:	70-90 minutes	
Set to light foot traffic:	after 24 hours	
Polishing:	after 3 days	
FINAL PERFORMANCE		
Adhesion (pull-out) wood-concrete (N/mm²):	2.5	
Adhesion (pull-out) wood-ceramic (N/mm²):	2.5	
Flexibility:	yes	
Shore A Hardness (after 7 days at +23°C):	80	

IMPORTANT NOTES

Whilst we try to ensure that any advice, recommendations or information given in our literature is accurate and correct, we have no control over the circumstances in which our product is used. It is therefore important that installers satisfy themselves that the product and conditions are suitable for the envisaged application. No warranty can be given or responsibility accepted other than, that the product supplied by us will meet our written specification. The installer should ensure that our latest product data and safety information sheets have been consulted prior to use.

Please refer to the current version of the Technical Data Sheet, available from our website www.mapei.com.

LEGAL NOTICE

The contents of this Technical Data Sheet ("TDS") may be copied into other project related documents, but the resulting document shall not supplement or replace requirements per the TDS in force at the time of the MAPEI product installation. The most up-to-date TDS can be downloaded from our website www.mapei.com.

ANY ALTERATIONS TO THE WORDING OR REQUIREMENTS CONTAINED OR DERIVED FROM THIS TDS EXCLUDES THE RESPONSIBILITY OF MAPEI.

All relevant references for the product are available upon request and from www.mapei.com.

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